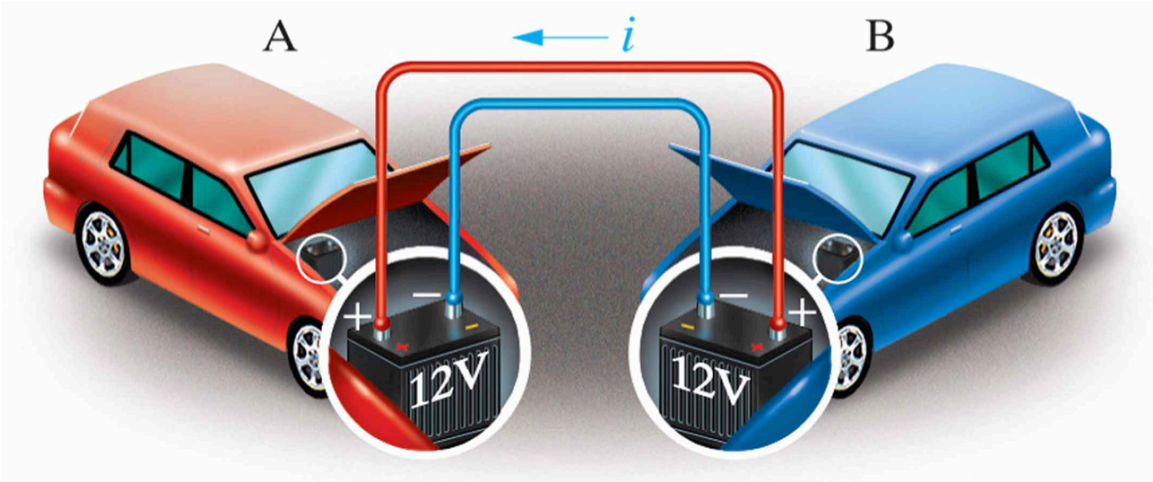
Problem # 1

When a car has a dead battery, it can often be started by connecting the battery from another car across its terminals. The positive terminals are connected together as are the negative terminals. The connection is illustrated in the following Figure. Assume the current *i* in the Figure is measured and found to be 30 A.

a) Which car has the dead battery?

b) If this connection is maintained for 1 min, how much energy is transferred to the dead battery?



Problem #2

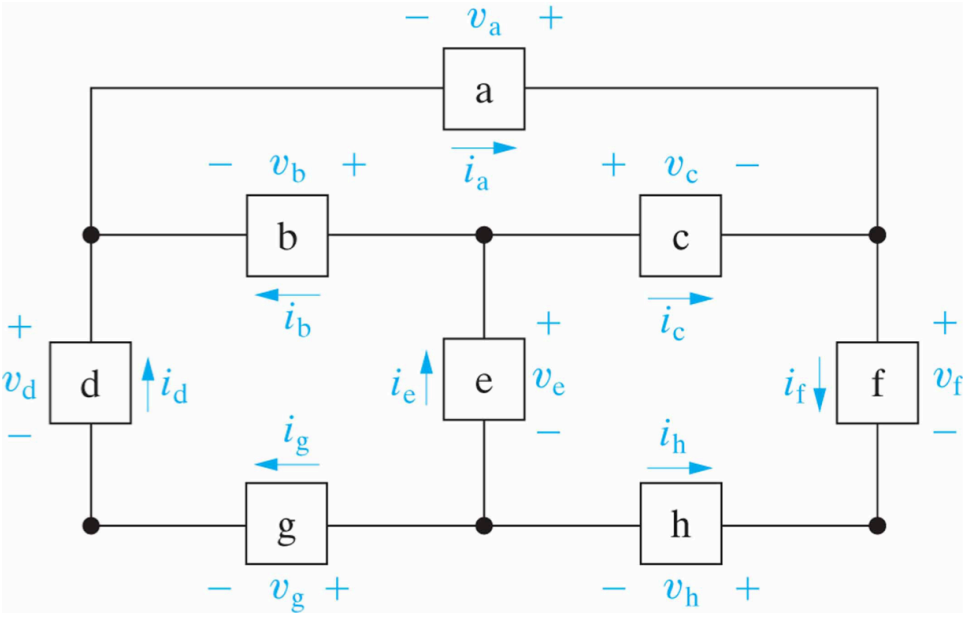
The manufacturer of a 1.5 V D flashlight battery says that the battery will deliver 9 mA for 40 continuous hours. During that time the voltage will drop from 1.5 V to 1.0 V. Assume the drop in voltage is linear with time. How much energy does the battery deliver in this 40 h interval?

Problem #3

Assume you are an engineer in charge of a project and one of your subordinate engineers reports that the interconnection in the given Figure does not pass the power check. The data for the interconnection are given in Table.

a) Is the subordinate correct? Explain your answer.

b) If the subordinate is correct, can you find the error in the data?



Element	Voltage (V)	Current (A)
a	46.16	6.0
b	14.16	4.72
c	−32.0	−6.4
d	22.0	1.28
e	33.6	1.68
f	66.0	−0.4
g	2.56	1.28
h	−0.4	0.4